

## **PROJECT DOCUMENTATION**

### **Project Title:**

### **Retail & E-commerce Sales Performance Analysis**

#### **1. Problem Statement**

Retail businesses generate large volumes of sales data every day, making it important to understand how different product categories, customer segments, and regions contribute to overall business performance. This project aims to analyze sales, profit, discounts, and product returns to identify trends, evaluate performance, and highlight areas that need improvement. The insights from this analysis will support better business decisions and help improve overall sales performance and profitability.

#### **2. Business Objectives**

- To clean and validate the Orders and Returns datasets.
- To transform the raw data into a structured and analysis-ready format.
- To analyze sales performance across product categories, customer segments, and regions.
- To identify the most profitable products and categories.
- To evaluate the impact of discounts on sales and profit.
- To analyze product return patterns.
- To build an interactive Power BI dashboard for business analysis.
- To provide data-driven business insights and recommendations for improving sales performance.

#### **3. Business Questions**

1. How are sales performing over time?
2. How does each product category perform in terms of sales, profit, orders, quantity sold, discounts, and returns?
3. Which sub-categories generate the highest sales and profit?
4. Which products generate the highest sales and profit?
5. Which brands contribute the highest sales and profit?
6. Which customer segments contribute the most sales and profit?
7. Which regions and states perform the best and which need improvement?
8. Which shipping modes are used most frequently, and how do they impact sales?
9. Which payment methods are preferred by customers?
10. How do discounts affect sales and profit?
11. Which products are returned most frequently, and what are the common return reasons?
12. What is the overall business performance based on key KPIs (Total Sales, Total Profit, Total Orders, Total Quantity Sold, Average Discount, Return Rate, and Profit Margin)?

#### **4. Scope**

The scope of this project includes collecting the Orders and Returns datasets, performing data profiling in Microsoft Excel, cleaning and transforming the data using Power Query, building a data model in Power BI, creating interactive dashboards, analyzing sales performance, and providing business insights and recommendations based on the analysis.

## 5. Deliverables

- Cleaned Orders dataset
- Cleaned Returns dataset
- Power Query transformation process
- Power BI data model
- Interactive dashboard
- Business insights
- Business recommendations
- Project documentation

## 6. Business Requirement Document

| Field            | Details                                                                                                                                     |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Stakeholders     | Business Management, Sales Team, Marketing Team, Data Analytics Team                                                                        |
| KPIs             | Total Sales, Total Profit, Total Orders, Total Quantity Sold, Average Discount, Return Rate                                                 |
| Assumptions      | The provided Orders and Returns datasets are complete except for identified data quality issues.                                            |
| Risks            | Missing values, duplicate records, inconsistent data, invalid values, and incorrect data types may affect analysis if not cleaned properly. |
| Success Criteria | Build a clean, accurate dataset and create an interactive dashboard that answers the business questions.                                    |
| Timeline         | Data Collection → Data Cleaning → Data Transformation → Data Modeling → Dashboard Development → Business Insights → Recommendations         |

## 7. Project Charter

| Field         | Details                                        |
|---------------|------------------------------------------------|
| Project Title | Retail & E-commerce Sales Performance Analysis |
| Duration      | 2-Weeks                                        |
| Tools Used    | Microsoft Excel, Power Query, Power BI         |